

Lenovo ThinkSystem ST250 V3

Enterprise server power for your
growing business



Lenovo

High Performance, High Value

The Lenovo ThinkSystem ST250 V3 tower server with a single Intel® Xeon® 6300-series/E-2400 processor and the ability to be installed in-rack or as a standalone server, it is optimized for growing businesses and remote applications. The ST250 V3 expands enterprise level compute for edge applications with the latest technology including Intel® Xeon® 6300-series/E-2400, DDR5 memory, and PCIe Gen5.

With support for up to 4x PCIe including one GPU the ST250 V3 has high-definition video output and accelerates complex workloads in remote installations like branch offices.

The ThinkSystem ST250 V3 also supports several workloads for business applications and is ideal for managing IT Infrastructure, and Edge Computing.

Flexible for Growth

The needs of growing businesses rapidly expand and change as they build momentum. The ThinkSystem ST250 V3 supports up to 4x PCIe slots, with one PCIe Gen5, and multiple storage types ranging from midsize to large capacity. With 8x 3.5" or 16x 2.5" HDD bays and up to 4x PCIe slots, the ST250 V3 is designed with the flexibility to manage the ever-expanding data that comes with growth.

The ST250 V3 can also be installed in a rack or as a standalone server making it ideal for managing business applications at the edge.

Remote Management made Easy

The Lenovo ThinkSystem ST250 V3 powered by the Intel® Xeon® 6300-series/E-2400 processor and XClarity system management, provides high performance while being easily managed. XClarity system management offers fast deployment and centralized management reducing the amount of onsite IT support required for operation. It can also be easily integrated into an existing network with other ThinkSystem servers.

The ThinkSystem ST250 V3 can be installed remotely while being centrally managed with XClarity, requiring only minimal onsite IT resources.

Secure and Reliable

The highly reliable ThinkSystem ST250 V3 is a part of Lenovo's x86 platform named #1 in reliability for 9 years in a row by ITIC. Boasting between 5 and 6 nines of reliability, customers can trust the ST250 V3 to always support their business applications.

Lenovo provides security throughout the entire lifecycle of the ST250 V3. Security starts at design with physical security features. Follows the server through the secure supply chain with hardware and firmware Root of Trust for secure deployment. ThinkShield and Intel provide data security features for secure operations of the server. At the end of the lifecycle, Lenovo offers Asset Recovery Services for secure disposal options.

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Specifications

Form Factor	4U chassis: Height: 430mm (16.9 inches), width: 175mm (6.9 inches), depth: 566mm (22.3 inches). Optional conversion to a rack-mount chassis
Processor	1x Intel® Xeon® 6300-series/E-2400 processor with up to 8 cores at 95W
Memory	Up to 128GB in 4x DIMM slots using 32GB UDIMMs 5600MHz TruDDR5
Drive Bay Combinations	4 or 8x 3.5" simple-swap SATA/SAS HDD (Up to 2x NVMe SSDs) 4 or 8x 3.5" hot-swap SATA/SAS HDD 8 or 16x 2.5" hot-swap SATA/SAS HDD Hybrid 4x 3.5" + 8x 2.5" hot-swap SATA/SAS HDD 2x 5.25" media bay (ODD/Tape) 1x Internal USB 3.2 G1 (5Gb) Optional 2x M.2 kit (SATA, in PCIe slots)
Expansion Slots	Support x8 Slimline with two PCH PCIe Gen4 x4 lanes PCIe x4 slot with PCH PCIe Gen4 x4 lanes PCIe x16 slot with CPU PCIe Gen5 x16 lanes PCIe x4 slot with PCH PCIe Gen4 x4 lanes PCIe x8 slot with PCH PCIe Gen4 x4 lanes
GPU Support	Optional NVIDIA Quadro T1000 or NVIDIA Quadro T400 GPU
HBA/RAID Support	Supports HW RAID adapters
Network Interface	2x 1GbE Embedded LOM 1x 1GbE Dedicated Management
Ports	Front: 1x USB 2.0 (XCC Mobile); 1x USB 3.2 Gen1 (5Gb) Rear: 4x USB 3.2 Gen1 (5Gb) ports; 2x RJ45 Ethernet port, 1x Dedicated Management port; 1x Serial port, 1x VGA port, NMI button
Systems Management and Security	Lenovo XClarity Controller 2; Root of Trust module with integrated Platform Firmware Resiliency, TPM 2.0 Optional front lockable bezel, Optional chassis intrusion switch
Power Supplies	Fixed 300W (Gold) / 500W (Titanium) Power Supply Unit (PSU) Redundant CRPS (Platinum) / CRPS (Titanium) PSU Fixed and Redundant PSU meets ErP lot 9
OS Support	Microsoft Windows Server, SUSE, Red Hat, VMware vSphere. Microsoft Windows 10/11 tested. Visit lenovopress.com/osig for more information.
Limited Warranty	1- and 3-year customer replaceable unit and onsite service, next business day 9x5, optional service upgrades

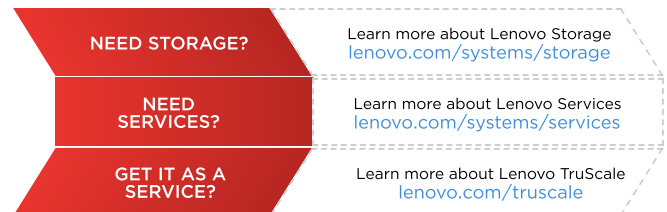


For More Information

To learn more about the Lenovo ThinkSystem ST250 V3, contact your Lenovo representative or Business Partner, or visit www.lenovo.com/thinksystem. For detailed specifications, consult the [ThinkSystem ST250 V3 product guide](#).

About Lenovo

Lenovo is a US\$83 billion revenue global technology powerhouse, ranked #153 in the Fortune Global 500, and serving millions of customers every day in 180 markets. Guided by its **vision** of “Smarter Technology for All”, Lenovo is executing a **Hybrid AI strategy** that spans Personal AI – one personal AI, multiple devices; and Enterprise AI – helping customers turn data into insights and value. This strategy is delivered through the Group’s commitment to **world-class innovation** and a full-stack AI portfolio, including devices (PCs, workstations, smartphones, tablets, accessories), infrastructure solutions (server, storage, edge, high performance computing and software defined infrastructure), as well as software, solutions, and services. With a global footprint spanning more than 20 research and development locations and a global supply chain that includes more than 30 manufacturing sites across 10 markets, Lenovo is widely recognized for its **operational excellence**. Lenovo is listed on the Hong Kong stock exchange under Lenovo Group Limited (HKSE: 992) (ADR: LNVGY). Learn more at www.lenovo.com and follow the latest news in our [newsroom](#).



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Lenovo ThinkSystem RAID 940 Series Internal RAID Adapters

Product Guide

The ThinkSystem RAID 940 family of PCIe 4.0 12 Gbps SAS RAID controllers are high-performance RAID-on-chip (ROC) server adapters for connectivity to drives internal to the server. These adapters support RAID levels 0, 1, 10, 5, 50, 6 and 60, and include an extensive list of RAS and management features.

Most RAID 940 adapters support Tri-Mode which enables the use of SAS, SATA and NVMe U.3 drives where each drive has a x1 connection to the adapter. On some servers, the 940-16i also supports an NVMe-only configuration, where each NVMe U.2 drive connects using a x4 connection.

The ThinkSystem RAID 940-16i is shown in the following figure.

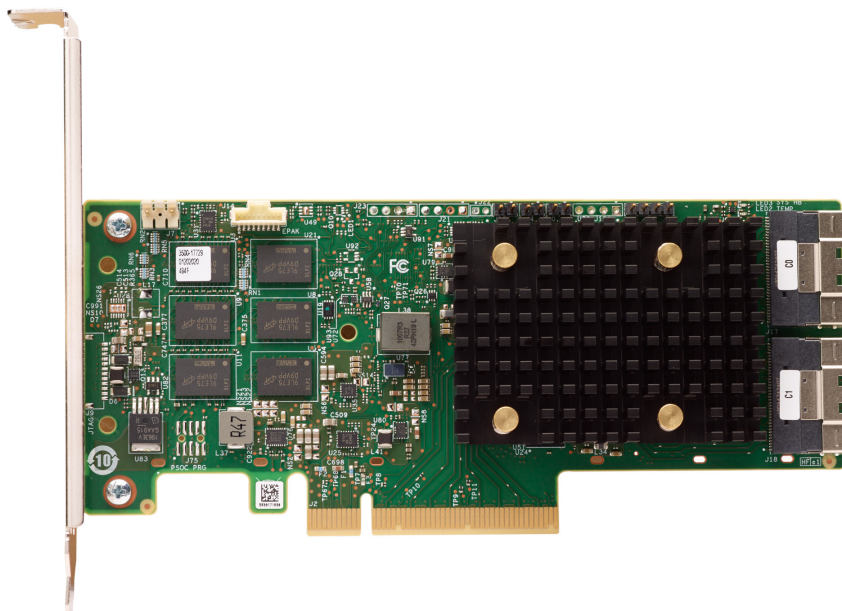


Figure 1. ThinkSystem RAID 940-16i Flash PCIe Gen4 12Gb Adapter

Did you know?

RAID on Chip-based controllers such as the RAID 940 adapters have a dedicated processor that offloads all RAID functions from the server's CPU. With hardware acceleration for RAID 5 and 6 operations plus up to 8 GB dedicated memory for caching, the 940-8i, 16i and 32i offer the ultimate performance for ThinkSystem servers.

Most of the RAID 940 adapters support NVMe through a feature named Tri-Mode. This feature enables the use of NVMe U.3 drives at the same time as SAS and SATA drives. NVMe drives connected using Tri-Mode support provide better performance than SAS or SATA drives, and NVMe drives typically also have lower latency and higher IOPS compared to SAS and SATA drives.

Part number information

The following table provides the ordering information for the RAID 940 adapters.

The table includes the three types of feature codes for CTO factory orders:

- SAS/SATA feature codes - the adapter is configured to support SAS or SATA drives only. No NVMe drive support.
- Tri-Mode feature codes - the adapter is configured to support SAS, SATA and NVMe U.3 drives. NVMe drives are connected using a x1 interface. NVMe U.2 drives are not supported.
- NVMe x4 feature codes - the adapter is configured to support NVMe U.2 drives. NVMe drives are connected using a x4 interface. SAS, SATA and NVMe U.3 drives are not supported.

For more information, see the [Features](#) section.

Table 1. Part numbers and feature codes

Part number	Feature codes			Description
	SAS/SATA	Tri-Mode	NVMe x4	
4Y37A09728	B8NY	BGM1	-	ThinkSystem RAID 940-8i 4GB Flash PCIe Gen4 12Gb Adapter
4Y37A09729	B8NW	BGM0	-	ThinkSystem RAID 940-8i 8GB Flash PCIe Gen4 12Gb Adapter
4Y37A78600	BM35	BM36	-	ThinkSystem RAID 940-16i 4GB Flash PCIe Gen4 12Gb Adapter
4Y37A09730	B8NZ	BDY4	BCYV	ThinkSystem RAID 940-16i 8GB Flash PCIe Gen4 12Gb Adapter
4Y37A09735	B8P0	BGM2	-	ThinkSystem RAID 940-16i 8GB Flash PCIe Gen4 12Gb Internal Adapter
4Y37A09733	B8P8	-	-	ThinkSystem RAID 940-32i 8GB Flash PCIe Gen4 12Gb Adapter

The ThinkSystem RAID 940-8i is shown in the following figure.

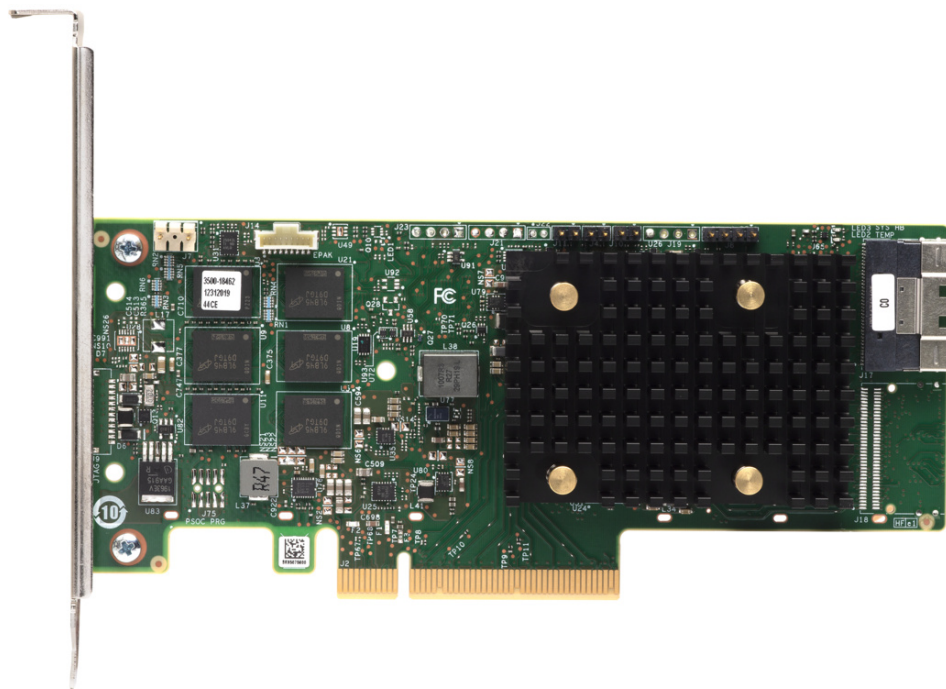


Figure 2. ThinkSystem RAID 940-8i 8GB Flash PCIe Gen4 12Gb Adapter

Technical specifications

The ThinkSystem RAID 940 Series internal RAID adapters have the following specifications:

- PCIe 4.0 x8 host interface
- Connectivity for up to 32 internal SAS or SATA drives:
 - RAID 940-8i supports up to eight internal drives
 - RAID 940-16i supports up to 16 internal drives
 - RAID 940-32i supports up to 32 internal drives
- 12 Gbps SAS/SATA RAID controllers, based on the Broadcom MegaRAID 9560 adapter family
- SAS/SATA support:
 - Supports 12Gb/s, 6Gb/s and 3Gb/s SAS (24 Gb/s drives operate at 12 Gb/s)
 - Supports 6Gb/s and 3Gb/s SATA
- Tri-Mode supported by all 8i (8-port) and 16i (16-port) adapters
 - Supports SAS drives
 - Supports SATA drives
 - Supports NVMe U.3 drives (x1 connection)
 - (No support for NVMe U.2 drives)
- NVMe x4 mode supported by 940-16i (limited server support)
 - Supports NVMe U.2 drives (x4 connection)
 - (No support for SAS, SATA or NVMe U.3 drives)
- 4GB or 8GB of integrated flash-backed cache
- Mixed drive support:
 - Support for intermixing of SAS and SATA drives
 - Tri-Mode: support intermixing of SAS, SATA and NVMe drives
 - Support for intermixing of HDDs and SSDs
 - Support for intermixing of drives of different speeds (for example, 12 Gb/s and 6 Gb/s)
 - Not supported: Mixing SAS and SATA drives in the same array
 - Not supported: Mixing of HDDs and SSDs in the same array
- Support for RAID 0, 1, 10, 5, 50, 6 and 60 standard
- Support for JBOD (non-RAID) drive state
- Support for up to 240 virtual drives per controller, 16 virtual drives per drive group
- Support for logical drive sizes greater than 2 TB
- Configurable stripe size up to 1 MB
- Supports 512e, 512n and 4K sector formatted drives
- Supports UNMAP command for SAS SSDs (JBOD and RAID volumes)
- Supports TRIM command for SATA SSDs (JBOD)
- Compliant with Disk Data Format (DDF) configuration on disk (CoD).
- S.M.A.R.T. support.
- Configuration through
 - XClarity Provisioning Manager UEFI interface
 - XClarity Controller web interface
 - XClarity Administrator Configuration Patterns
 - StorCLI command-line interface
 - LSI Storage Authority (LSA) GUI interface
 - UEFI Human Interface Infrastructure (HII)